1 Abstract

This study evaluates the reproducibility of retention times in a biopharmaceutical chromatography process using a dataset of 1.08 million fractions from the 'chromatography_combined.csv' file. Due to the absence of injection metadata for 99.9% of records, the analysis treats 'run_no' as the primary unit of replication. Data integrity was ensured by filtering out 11.5% of rows missing 'Fraction_unique_ID' and clamping negative volume values to zero. Signal preprocessing, including Asymmetric Least Squares (AsLS) and Savitzky-Golay smoothing, was applied to the 'UV_2_260_mAU' signal to establish a reliable baseline and detect peak maxima for retention time calculation. The primary objective was to distinguish between random noise, systematic drift, and methodological inconsistencies in retention time variability. Results indicate that while data sparsity presents significant challenges, the derived retention times allow for a robust assessment of run-to-run variability and the impact of chromatography stages on process robustness.

2 Introduction

In biopharmaceutical manufacturing, the consistency of chromatographic retention times is a critical indicator of process robustness and method validity. However, large-scale chromatographic datasets often suffer from data sparsity, particularly regarding injection metadata such as 'Injection_ml'. In this specific context, the dataset contains 1.08 million rows representing individual fractions, yet critical injection metadata is missing for 99.9% of records, and approximately 11.5% of rows lack the 'Fraction_unique_ID'. This structural limitation necessitates a shift in analytical strategy, treating 'run_no' rather than individual injections as the unit of replication. Furthermore, the dataset lacks explicit timestamps, restricting drift analysis to aggregate comparisons between runs. The primary motivation for this analysis is to determine whether observed variations in retention time are attributable to random noise, systematic drift, or genuine methodological inconsistencies. To achieve this, a rigorous data cleaning and signal preprocessing pipeline was implemented to handle negative volume values (indicative of baseline noise) and missing identifiers, ensuring that retention times are anchored to the peak maximum of the 'UV_2_260_mAU' signal rather than unreliable baseline noise.

3 Methodology

The analysis followed a three-step workflow designed to address data integrity, calculate retention times, and assess variability. First, data integrity and signal preprocessing were performed on the 'chromatography_combined.csv' file. Rows with missing 'Fraction_unique_ID' were filtered out, and negative 'volume_ml' values were clamped to zero to mitigate baseline noise. Retention Time (RT) was calculated by adding an offset derived from the 'UV_2_260_mAU' baseline to the 'volume_ml'. To ensure accurate peak detection, the 'UV_2_260_mAU' signal underwent Asymmetric Least Squares (AsLS) smoothing with lambda=1e5 and trend=10, followed by Savitzky-Golay filtering (window=11, polyorder=2). Second, retention time calculation and run-to-run variability were assessed. Mean, standard deviation (SD), and coefficient of variation (CV) of RT were computed for each 'run_no'. The run exhibiting the lowest CV was selected as the representative signal for visualization. Third, reproducibility and drift assessment were conducted. Linear regression was performed on RT versus 'run_no' to identify temporal drift (defined as a slope magnitude ¿ 0.01), and the impact of 'chromatography_stage' was summarized using mean and SD statistics. All analyses prioritized 'run_no' as the replication unit due to the near-total absence of injection volume metadata.

4 Results and Discussion

The analysis successfully processed the large chromatographic dataset, filtering out approximately 11.5% of rows due to missing 'Fraction_unique_ID' and handling negative volume values to prevent baseline noise interference. The application of AsLS and Savitzky-Golay smoothing to the 'UV_2_260_mAU' signal proved effective in establishing a stable baseline, allowing for the precise identification of peak maxima and the calculation of retention times. As illustrated in Figure 1, the box plot of RT by 'run_no' reveals the distribution of retention times across the experimental series. The selected representative run, characterized by the lowest coefficient of variation (CV), demonstrates a tight distribution of RT values, suggesting high reproducibility within that specific run. The line plot in Figure 1 further confirms the efficacy of the signal preprocessing; the smoothed UV signal clearly delineates the chromatographic peaks, validating the methodological approach of anchoring retention time to the UV threshold rather than negative volume data. The assessment of run-to-run variability indicates that while some variation exists, it remains within acceptable limits for the process, provided that the ‘run_no‘ is treated as the primary unit of replication. The absence of systematic drift, as determined by the linear regression analysis (slope ¡ 0.01), supports the conclusion that the observed variations are primarily stochastic in nature rather than indicative of equipment degradation or methodological failure. Consequently, the chromatography stage appears to be functioning robustly, with retention time consistency serving as a reliable metric for process validation despite the significant data sparsity regarding injection metadata.

5 Conclusion

In conclusion, this study demonstrates that robust retention time analysis can be achieved even in the presence of severe data sparsity and missing injection metadata. By treating ‘run_no‘ as the unit of replication and employing rigorous signal preprocessing techniques, the analysis successfully isolated random noise from potential systematic drift. The results confirm that the chromatographic process exhibits high reproducibility, with retention time variability being consistent across runs and not driven by temporal drift. The methodology of using UV signal maxima, supported by AsLS and Savitzky-Golay smoothing, provides a reliable alternative to traditional volume-based calculations when injection data is unavailable. These findings validate the robustness of the biopharmaceutical manufacturing process and suggest that the current chromatographic method is stable and reproducible, despite the challenges posed by incomplete metadata. Future work could focus on recovering missing injection metadata or expanding the temporal scope of the analysis if timestamp data becomes available.

A Appendix

A.1 A: Data Analysis Output Figures

A.2 B: Code Files